

Tour Length Comparison - 100-City Maps

Only tour length is shown. Source 100-200 values are sorted within each panel by BlindBlendACO tour length: 1-101.

Algorithm colors

BaselineACO

BlindBlendACO

ElitistACO

MaxMinACO

Tour length averages

With Cross Check

BaselineACO: 16,010

BlindBlendACO: 15,623

ElitistACO: 15,941

MaxMinACO: 15,710

Without Cross Check

BaselineACO: 16,364

BlindBlendACO: 15,826

ElitistACO: 16,182

MaxMinACO: 15,973

With Cross Check

This line chart displays the tour length for four algorithms across 101 sorted maps/runs. The y-axis, labeled 'Tour length', ranges from 14,332 to 18,618. The x-axis, labeled 'Sorted map / run index', ranges from 1 to 101. Four data series are plotted: BaselineACO (blue), BlindBlendACO (purple), ElitistACO (red), and MaxMinACO (green). Each series includes a dashed horizontal line representing its average tour length. BaselineACO has the highest average (16,010), followed by ElitistACO (15,941), MaxMinACO (15,710), and BlindBlendACO (15,623). The chart shows significant fluctuations in tour length across the sorted maps, with BaselineACO generally performing the worst and BlindBlendACO the best.

Without Cross Check

This line chart displays the tour length for the same four algorithms across 101 sorted maps/runs, but without a cross-check. The axes and data series are identical to the 'With Cross Check' chart. However, the overall tour lengths are higher, reflecting the absence of the cross-check optimization. The average tour lengths are: BaselineACO (16,364), ElitistACO (16,182), MaxMinACO (15,973), and BlindBlendACO (15,826). The relative performance of the algorithms remains consistent, with BaselineACO being the least efficient and BlindBlendACO the most efficient.

Sorted map / run index